

Standard Operating Procedure (SOP) for GenoToxMass

1. Scope and Purpose

This Standard Operating Procedure (SOP) defines the complete workflow for predicting chemical genotoxicity using the GenoToxMass platform, from sample preparation to result interpretation. This protocol is designed to enable regulatory agencies, contract research organizations, and analytical laboratories to independently validate and implement the method under standardized conditions.

2. Instrumentation and Materials

2.1 Chromatographic System

- Recommended Column: C18 column (2.1-3.5 μm particle size, 2.1 $\times$ 50 mm dimensions)
- Column Temperature: 30 $\pm$ 5 $^{\circ}\text{C}$
- Mobile Phase A: Water with 0.1% formic acid (LC-MS grade)
- Mobile Phase B: Acetonitrile with 0.1% formic acid (LC-MS grade)
- Flow Rate: 0.3-1.0 mL/min
- Injection Volume: 1-5 μL
- Gradient Program:

Time (min)	%B
0.0	5
1.0	5
8.0	95
10.0	95
10.1	5
12.0	5

2.2 Mass Spectrometry System

- Ionization Source: Electrospray Ionization (ESI)
- Scanning Method: Select the mode exhibiting higher signal strength
- Capillary Voltage: 5.0 kV
- Ion Source Temperature: 325 $^{\circ}\text{C}$
- Desolvation Gas Flow: 12 L/min (N_2)
- Cone Gas Flow: 12 L/min (N_2)
- Collision Energy: Ensure the molecular ion peak is retained at moderate intensity (10%-80%)
- Mass Range: m/z 50-1000
- Resolution: $\geq 20,000$ (FWHM) at m/z 200

2.3 Quality Control Standards

- System Suitability Test Mixture: Colchicine (CAS 64-86-8), Reserpine (CAS 50-55-5) at 1 $\mu\text{g/mL}$ each
- Acceptance Criteria: Retention time variation ≤ 0.2 min; mass accuracy ≤ 5 ppm; intensity variation $\leq 20\%$ across three consecutive injections

3 Sample Preparation

3.1 Environmental Water Samples

- (1) Filter sample through 0.22 μm nylon membrane filter
- (2) Add internal standard at 100 ng/mL final concentration
- (3) Concentrate 100 mL sample using solid-phase extraction (SPE)
- (4) Condition SPE cartridge with 5 mL methanol, then 5 mL water
- (5) Load sample at 5 mL/min

- (6) Wash with 5 mL 5% methanol in water
- (7) Elute with 5 mL methanol
- (8) Evaporate to dryness under N₂ at 40°C
- (9) Reconstitute in 200 µL methanol:water (1:1, v/v)

3.2 Pharmaceutical Impurities

- (1) Dissolve drug substance in appropriate solvent at 10 mg/mL
- (2) Dilute to 10 µg/mL with methanol:water (1:1, v/v)
- (3) Filter through 0.22 µm PTFE syringe filter

3.3 Food Matrices

- (1) Homogenize sample (5 g) with 10 mL acetonitrile
- (2) Centrifuge at 10,000 × g for 10 min at 4°C
- (3) Collect supernatant and perform dispersive SPE cleanup
- (4) Evaporate 5 mL supernatant to dryness
- (5) Reconstitute in 500 µL methanol:water (1:1, v/v)

4 Data Acquisition

4.1 Acquisition Mode

- Use data-dependent acquisition (DDA) or data-independent acquisition (DIA)
- For completely unknown samples, both positive and negative ionization modes need to be acquired, and the spectra from the mode with higher overall intensity should be used for subsequent analysis
- Acquire MS/MS spectra
- Dynamic exclusion: 5 s after two occurrences

4.2 File Format Requirements

- Export raw data as .raw
- Alternatively, use the GenoToxMass template (.xlsx) with the following required columns:
 - (1) *mz_list*: Comma-separated list of *m/z* values (e.g., “121.05, 149.02, 177.05”)
 - (2) *intensity_list*: Comma-separated list of corresponding intensities (e.g., “100, 45, 12”)

5 Data Preprocessing and Feature Extraction (Software automatic processing)

5.1 Spectral Filtering

- (1) Identify the maximum intensity (MaxM) in the raw spectrum
- (2) Set filtering threshold $T = \text{MaxM} \times P$, where $P = 0.01$ (1%)
- (3) Retain only peaks with intensity $\geq T$

5.2 Feature Calculation (14 Parameters)

Feature	Abbreviation	Calculation Method
Peak Number	PN	Count of peaks after filtering
Base Peak	BP	<i>m/z</i> value of the most intense peak in filtered spectrum
Base Peak Proximity	BPP	The <i>m/z</i> difference between the BP and its nearest peak
Maximum Mass	MaxM	Maximum <i>m/z</i> value in filtered spectrum
Maximum Mass Proximity	MaxMP	The <i>m/z</i> difference between the MaxM and its nearest peak
Minimum Mass	MinM	Minimum <i>m/z</i> value in filtered spectrum

Mass Mean	MM	Mean of all m/z values in filtered spectrum
Mass Standard Deviation	MSD	Standard deviation of all m/z values
Intensity Mean	IM	Mean of all intensity values
Intensity Standard Deviation	ISD	Standard deviation of all intensity values
Intensity Density	ID	BPI / PN
Retention Time	RT	As acquired (minutes)
Collision Energy	CE	As acquired (eV)
Precursor Type	PT	0 for $[M-H]^-$, 1 for $[M+H]^+$, $[M+Na]^+$, $[M+K]^+$

* If user prefer not to input the three test-related parameters (RT, CE, and PT), a generic input profile with RT fixed at 8 min, CE at 30 eV, and PT set to positive ion mode may be applied. However, it is important to note that if user select a specific ionization mode, the spectra must be acquired under that same mode. This approach is not recommended unless absolutely necessary, as the predictive performance under this mode is expected to degrade by $\geq 10\%$ relative to the external validation results.

6 Model Invocation

6.1 Online Platform Access

- (1) Navigate to <https://mspredict.com/genotoxic>
- (2) Download the input template from the website
- (3) Prepare your data following the template format
- (4) Upload the completed .xlsx file
- (5) Click “Predict” and wait for processing (typically < 5 seconds)
- (6) Results will display as:
 - Prediction: “Toxic” or “Non-Toxic”
 - Probability: Value between 0 and 1

7 Result Interpretation and Decision Framework

7.1 Default Classification Threshold

- Optimal threshold τ^* (determined by user need)
- Classification rule:
 - If probability $\geq \tau^* \rightarrow$ Classify as Toxic
 - If probability $< \tau^* \rightarrow$ Classify as Non-Toxic

7.2 Threshold Adjustment for Specific Applications

Application Priority	Recommended Threshold	Expected Performance Trade-off
Maximize Accuracy	0.42	Balanced sensitivity/specificity
Minimize False Negatives (Safety-First)	0.30	Higher recall, lower precision
Minimize False Positives (Cost-Sensitive)	0.65	Higher specificity, lower recall
Regulatory Confirmation	0.55	Optimized for both

7.3 Confidence Scoring

- High Confidence: Probability ≥ 0.8 or ≤ 0.2
- Medium Confidence: $0.6 \leq \text{Probability} < 0.8$ or $0.2 < \text{Probability} \leq 0.4$
- Low Confidence: $0.4 < \text{Probability} < 0.6$

8 Quality Control and Validation

8.1 System Suitability Test

- (1) Inject system suitability test mixture (colchicine, reserpine)
- (2) Verify retention times within ± 0.2 min of established values
- (3) Verify mass accuracy ≤ 5 ppm for all three compounds
- (4) Run GenoToxMass prediction on all three
- (5) If any prediction deviates, investigate instrument performance

8.2 Batch Quality Control

- Include one positive control (e.g., colchicine) and one negative control (e.g., reserpine) in each analytical batch
- Acceptance criteria: correct classification for both controls
- If controls fail, reject the entire batch and re-analyze

8.3 Inter-Laboratory Transfer

- For laboratories implementing this method for the first time:
 - (1) Analyze the reference standard set (n=20 compounds)
 - (2) Compare predictions with reference values
 - (3) Acceptable performance: $\geq 85\%$ concordance
 - (4) Document all results for regulatory audit

9 Documentation and Reporting

All analyses should be documented with:

- (1) Instrument calibration records
- (2) System suitability test results
- (3) Sample preparation logs
- (4) Raw data files
- (5) Feature extraction worksheets
- (6) Model prediction outputs
- (7) Final interpretation and decision rationale

10 Troubleshooting Guide

Problem	Possible Cause	Solution
Low peak count after filtering	High noise level	Increase injection volume; improve sample cleanup
Unexpected predictions	Poor spectral quality	Verify collision energy; check precursor ion selection
Platform unavailable	Server maintenance	Contact support